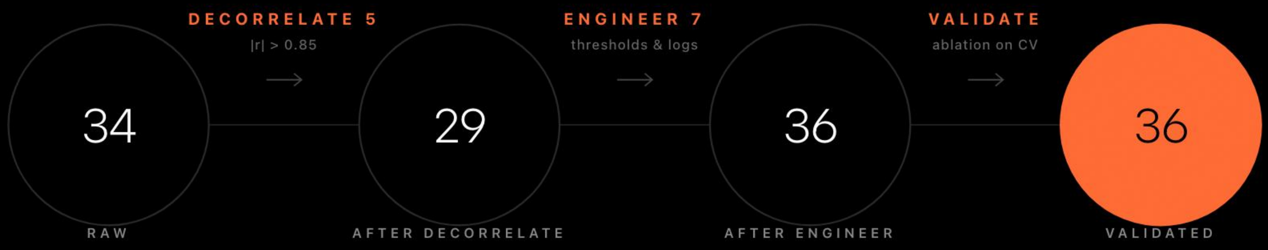
WIDS GLOBALTHON · WILDFIRE RISK

Better features.
Better model.

THE RECIPE

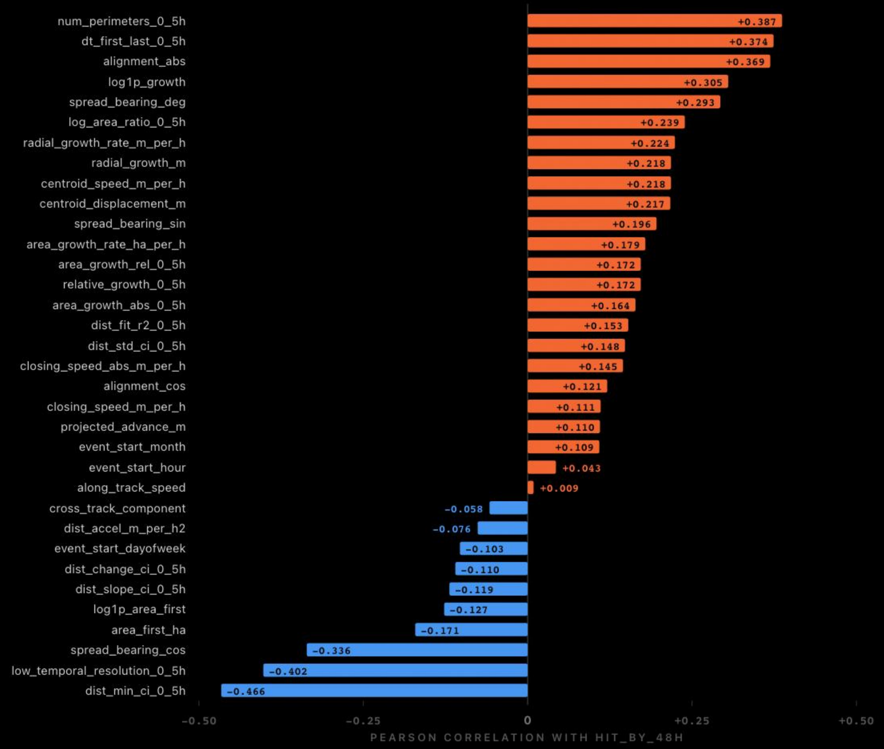
Screen. Decorrelate. Engineer. Validate.



STEP 1 SCREEN

First, see what speaks to the target.

All 34 features by Pearson correlation with `hit_by_48h`. Orange = positive (more likely to hit) · Blue = negative (less likely). A linear screen — low correlation isn't a verdict, since trees can still find non-linear signal.



STEP 2A FIND DUPLICATES

Some signals are the *same* signal.

Across the 34-feature matrix, **25 pairs** exceed `|r| > 0.85`.

PEARSON CORRELATION COEFFICIENT

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \equiv \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}$$

−1 ——— 0 ——— +1

PERFECT INVERSE · NO LINEAR · PERFECT DIRECT

$|r| > 0.85$ ⇒ two features carry essentially the same information

	area_growth_abs	area_growth_rate	log_area_ratio	radial_growth_m	radial_growth_rate	centroid_displacement	centroid_speed	dist_std
area_growth_abs	1.00							
area_growth_rate	0.99	1.00						
log_area_ratio			1.00					
radial_growth_m	0.87	0.91		1.00				
radial_growth_rate			0.93	0.97	1.00			
centroid_displacement		0.90	0.85	0.99	0.97	1.00		
centroid_speed			0.92	0.94	0.99	0.96	1.00	
dist_std	0.91	0.88						1.00

PEARSON CORRELATION

0.70 — 1.00

OUTSIDE THIS CLUSTER

1.000

`relative_growth_0_5h`
≡
`area_growth_rel_0_5h`

TWO FEATURES. ONE SIGNAL.

STEP 2B DECORRELATE

Why these five are dropped.

Each row: the dropped feature, its near-perfect twin (or "near-constant" if it had none). The model needs one feature per redundant pair, not both. 34 → 29.

	DROPPED	REASON	KEPT TWIN
01	~~closing_speed_m_per_h~~	r = −0.998 REDUNDANT	→ dist_change_ci_0_5h
02	~~closing_speed_abs_m_per_h~~	r = +0.907 REDUNDANT	→ area_growth_abs_0_5h
03	~~dist_std_ci_0_5h~~	r = −0.943 REDUNDANT	→ dist_slope_ci_0_5h
04	~~projected_advance_m~~	r = +0.855 REDUNDANT + SPARSE	→ area_growth_abs_0_5h
05	~~dist_fit_r2_0_5h~~	near-constant NEAR-CONSTANT	—

PER-HORIZON BRIER IMPACT · base_full → after manual cut

12h	24h	48h	72h
−0.034 HELPED	+0.014 HURT	−0.001 HELPED	−0.001 HELPED

5 features removed · 4 had a near-perfect twin · 1 was nearly constant

34 → 29

STEP 3 ENGINEER

Build what's missing.

The raw feature `dist_min_ci_0_5h` is the minimum fire-to-target distance observed in the first 5 hours — by far the strongest single signal in the data. Two examples below show how transforming it lifts the model's importance score even further. 29 → 36.

RAW FEATURE

`dist_min_ci_0_5h`

IMPORTANCE

12.77

THRESHOLD FLAG →

`(dist_min < 5000).astype(int)`

★ ENGINEERED

`dist_close`

IMPORTANCE

59.20

×4.6

LIFT

WHY IT WORKS All 66 hits sit under 5 km · 0 of 152 misses do.

RAW FEATURE

`dist_min_ci_0_5h`

IMPORTANCE

12.77

LOG COMPRESSION →

`np.log1p(dist_min)`

★ ENGINEERED

`log_dist_min`

IMPORTANCE

14.33

×1.1

LIFT

WHY IT WORKS Linear scale buries the signal in the tail. Log restores it.

STEP 4 VALIDATE

Which feature set should we ship?

Three candidate sets — raw 34, after the manual cut (29), after engineering (36) — scored on CatBoost validation Brier (lower is better). The engineered set wins decisively on 48h and 72h, where Brier drops 4–5× vs. the raw set.

	HORIZON 12h	HORIZON 24h	HORIZON 48h	HORIZON 72h
34 · RAW ALL FEATURES	0.123	BEST 0.047	0.010	0.005
29 · + MANUAL CUT −5 REDUNDANT (SLIDE 5)	BEST 0.089	0.061	0.009	0.004
36 · + ENGINEERED +7 TRANSFORMS (SLIDE 6)	0.097	0.059	BEST 0.002	BEST 0.001

CatBoost validation Brier · lower is better

→ SHIP THE 36-FEATURE SET

THREE RULES

01 Cut what's *redundant*.

02 Build what's *missing*.

03 Trust *validation*.

BETTER FEATURES. BETTER MODEL.

34 → 29 → 36 · AUC 0.9987 at 48h